Machine Learning

02

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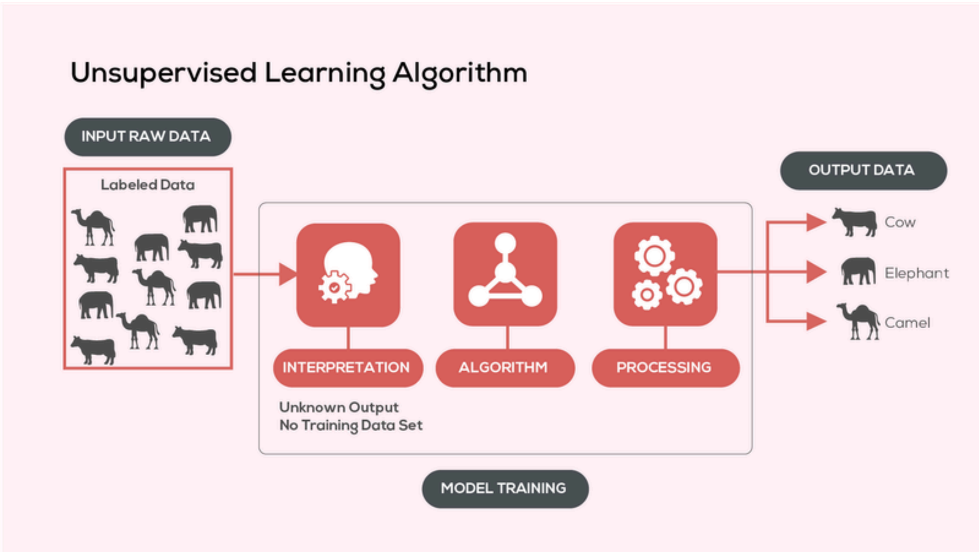
Unsupervised Learning

Unsupervised Learning is a type of machine learning where the model works without labelled data. It learns patterns on its own by grouping similar data points or finding hidden structures without any human intervention.

- It is used for tasks like clustering, dimensionality reduction and Association Rule Learning.
- Helps identify hidden patterns in data
- Useful for grouping, compression and anomaly detection



Working of Unsupervised Learning

Unsupervised learning works with data that has no labels, so the model tries to find patterns on its own.

1. Collect Data

We start with data that doesn't have any categories or answers.

Example: Photos of animals without names.

2. Choose an Algorithm

We pick a method based on what we want to do.

- Group similar data → Clustering (K-Means)
- Find patterns → Association rules
- Reduce complexity → PCA

3. Train the Model
We give all the data to the model.
The model starts finding similarities and relationships by itself.

4. Find Patterns or Groups
The model automatically:
- Groups similar items together
- Finds hidden patterns
- Simplifies complex data

Example: It might group similar animals together
.
5. Understand the Results
We analyze what the model found and use it for:
- Insights
- Visualization
- Detecting unusual data

Types

Clustering Algorithms

Clustering is an unsupervised learning technique that groups similar data together without any labels.
- Groups data with similar features
- Helps find patterns in raw data
- Used in things like customer segmentation and anomaly detection
- Works without knowing the correct answers

Common Clustering Methods
- K-Means → Groups data into K clusters based on similarity
- Hierarchical → Builds clusters step-by-step like a tree
- DBSCAN → Finds dense groups and ignores noise
- Mean-Shift → Moves data toward dense areas to form clusters
- Spectral → Uses connections between data points to group them

Association Rule Learning

Association rule learning is an unsupervised technique used to find relationships between items in data.
It works using "if–then" rules
Example: If a person buys bread → they also buy butter

Key Points

- Finds items that often occur together
- Used in market basket analysis
- Helps in promotions and product recommendations
- Works without labeled data

Common Algorithms

- Apriori → Finds frequent item combinations step-by-step
- FP-Growth → Faster method, avoids unnecessary steps
- Eclat → Uses itemset intersections for efficiency
- Tree-based methods → Handles large datasets easily

Dimensionality Reduction

TDimensionality reduction means reducing the number of features in a dataset while keeping important information.
It makes data:

- Easier to understand
- Faster to process
- Less noisy

Key Points

- Reduces number of variables
- Keeps only important features
- Improves model performance
- Helps avoid overfitting

Common Algorithms

- PCA → Converts data into important components
- NMF → Breaks data into simpler non-negative parts
- LLE → Keeps relationships between nearby points
- Isomap → Preserves overall structure of data